AILT9018 Module 3: AI as Professional Assistant

Technical Reading with AI

Read Faster, Verify Smarter

Your Mission

Use AI to investigate a technical document without letting the AI do your engineering thinking for you.

In 90 minutes you will: extract facts, demand evidence, challenge uncertain claims, and keep humans accountable.

What You Will Produce

1. A structured technical summary
2. A source-linked requirement table
3. A hallucination and risk tracker
4. An improved technical-reading prompt
5. An AI-Assisted Technical Reading Memo
6. An Ed Discussion post and peer response

AI Platform: Use HKU GenAI for all AI tasks: `https://genai.hku.hk`

Learning Outcomes

By the end of this workshop you will be able to:

1. Use HKU GenAI to generate a structured technical summary.
2. Extract technical requirements with section references and evidence.
3. Write stronger prompts for technical-reading tasks.
4. Verify AI-generated claims against source evidence.
5. Identify at least three risks of AI-assisted technical reading.
6. Produce a short technical reading memo.
7. Explain an engineer's ethical responsibility when using AI.

Schedule Overview

Time	Activity	Mode	Output	Submit?
0–5	Launch & AI Checklist	Individual	Completed checklist	Keep
5–15	Mission 1: Skim Like an Engineer	Individual → Pair	Human vs AI summaries	Pair compare
15–27	Mission 2: Prompt Upgrade	Pair	Weak vs strong prompts	Prompt battle
27–47	Mission 3: Requirement Treasure Hunt	Group of 3	Requirement table	Group check
47–62	Mission 4: Hallucination Hunt	Pair	Verified claim tracker	Pair challenge
62–70	Mission 5: Engineer's Reality Check	Group of 3–4	Three risks, one safeguard	Class share
70–80	Mission 6: Ed Discussion Exchange	Individual → Peer	Post and reply	Submit
80–87	Mission 7: Reading Memo Sprint	Individual	Technical reading memo	Keep/share
87–90	Exit Ticket	Individual	Three reflections	Submit

Active working time: 85 of 90 minutes (94%).

Materials Needed

- Laptop with internet access
- HKU GenAI: https://genai.hku.hk
- Ed Discussion access
- The fictional technical document (Section 10)
- Workbook and templates (Sections 7 and 11)
- A partner ready to challenge your AI output

Responsible AI Checklist

Complete this **before** opening the AI platform.

☐ I will use https://genai.hku.hk for GenAI tasks.

☐ I will not upload sensitive, confidential, personal, or unpublished data.

☐ I will verify AI outputs against the source document.

☐ I will not treat AI as an engineering authority.

☐ I will cite section or page references in AI-assisted summaries.

☐ I understand that an AI summary is not evidence unless traceable to the source.

☐ I will check units, limits, warnings, and conditions manually.

Activity	CLO 1: Use AI tools	CLO 3: Ethics & limits	CLO 2: Systems
Human vs AI Skim	✓	✓	
Prompt Upgrade	✓	✓	
Requirement Treasure Hunt	✓	✓	✓
Hallucination Hunt	✓	✓	✓
Engineer's Reality Check		✓	
Technical Reading Memo	✓	✓	✓

☐ I understand that engineers remain accountable for technical decisions.

Professional Reality Check

AI can help you read faster. It cannot accept legal, ethical, or professional responsibility for your decisions. You can.

Activity Plan

0–5 min: Launch

1. Open this workbook.
2. Complete the Responsible AI Checklist.
3. Open the technical document (Section 10).
4. Open HKU GenAI but do not enter the document yet.
5. Pair up with the person next to you.

5–15 min: Mission 1 — Skim Like an Engineer

Challenge: Can AI spot more than you in two minutes?

1. Skim Sections 1–3 for two minutes. Write down:
   - The system's purpose
   - Two requirements
   - One risk
2. Give the same text to HKU GenAI using the quick-summary prompt.
3. Compare your findings with the AI output. Circle:
   - One useful AI finding
   - One vague or questionable AI claim
   - One important point the AI missed

15–27 min: Mission 2 — Prompt Upgrade

Challenge: Turn a lazy prompt into an engineering-grade instruction.

1. Run the weak prompt: *"Summarise this document."*
2. Record what is missing from the output.

3. Run the strong prompt (see workbook).
4. Compare outputs on four criteria: structure, source evidence, handling of uncertainty, engineering usefulness.
5. Swap prompts with another pair. Improve their prompt by adding one of:
   - Evidence requirement
   - Output format
   - Uncertainty instruction
   - Safety check

27–47 min: Mission 3 — Requirement Treasure Hunt

Challenge: Extract facts without inventing details.
Assign roles in groups of three:

AI Operator enters prompts

Source Inspector checks the document

Sceptical Engineer challenges unclear claims

1. Ask the AI to extract requirements, constraints, numerical parameters, assumptions, and risks.
2. Require a section reference and evidence quote for every item.
3. Enter at least five items in the requirement table.
4. Manually verify each item.
5. Mark anything ambiguous, incomplete, or unsupported.
6. Rotate roles after ten minutes.

47–62 min: Mission 4 — Hallucination Hunt

Challenge: Catch the AI acting confident without evidence.

1. Ask the AI five targeted questions from the workbook.
2. Add one trick question about information **not** in the document.
3. For every answer, find supporting evidence.
4. Label each answer: Supported / Partially Supported / Unsupported / Contradicted.
5. Score: 1 point per unsupported claim caught, 2 points per contradiction, 1 bonus for invented details.

Professional Reality Check

If this were a real installation, flood warning system, or medical device—what would you verify before acting?

62–70 min: Mission 5 — Engineer's Reality Check

Discuss in groups of 3–4:

1. One useful role for AI in technical reading?
2. One task you would not delegate to AI?
3. What could happen if AI misses a warning or unit?
4. Who is accountable if AI interpretation leads to a design error?
5. What information must never be uploaded?

Record: three risks, one safeguard, one professional-responsibility statement.

70–80 min: Mission 6 — Ed Discussion Exchange

Part A (5 min): Post your best prompt, what you changed, and one likely AI failure.
Part B (5 min): Reply to one classmate with a verification suggestion.

80–87 min: Mission 7 — Reading Memo Sprint

Use your verified evidence to complete the AI-Assisted Technical Reading Memo template. Do not copy an AI summary directly—you are responsible for checking every statement.

87–90 min: Exit Ticket

Complete three statements (see Section 13).

Student Workbook

Mission 1: Skim Like an Engineer

Step 1: Human Skim (2 minutes)

- The system is designed to: ____________________
- Two requirements I spotted:
  1. ____________________
  2. ____________________
- One possible risk: ____________________

Step 2: AI-Assisted Skim

Prompt

Analyse Sections 1–3 of the document. In no more than 120 words, state the system purpose, three explicit requirements, two numerical parameters, and one safety concern. For every claim, provide the source section and a short evidence quote. If information is absent, write "not specified". Do not infer missing details.

Step 3: Compare

Comparison	Your note
Useful point found by AI	
Important point missed by AI	
Vague or unsupported AI claim	
Point you found faster than AI	

Pair discussion: Which reader was more useful—you, the AI, or the combination?

Mission 2: Prompt Upgrade

Weak Prompt

"Summarise this document."

What does it fail to specify?

- ☐ Purpose
- ☐ Output format
- ☐ Source evidence
- ☐ Treatment of missing information
- ☐ Safety focus
- ☐ Level of detail
- ☐ Units and conditions

Strong Prompt

Prompt

Act as a careful engineering document assistant. Analyse only the supplied document. Produce:

1. A three-sentence purpose summary
2. A table of explicit requirements and constraints
3. All numerical parameters with units and operating conditions
4. Assumptions, ambiguities, warnings, and missing information

For each item, cite the section and quote the shortest relevant evidence. Separate explicit statements from interpretations. Write "not specified" when evidence is absent. Do not invent standards, test results, or design recommendations.

Prompt Battle Scorecard (0–2 each)

Criterion	Weak	Strong
Requirements clearly separated		
Section references included		
Evidence quotes accurate		
Missing information labelled		
Safety concerns visible		
Total / 10		

Prompt Upgrade Template

Analyse [**document/sections**] for the purpose of [**task**]. Extract [**information needed**]. Present as [**format**]. For each claim, give [**section/evidence**]. Separate [**facts**] from [**inferences**]. If information is missing, [**required response**]. Do not [**prohibited behaviour**].

Mission 3: Requirement Treasure Hunt

AI Prompt

Extract all explicit engineering requirements and constraints from the supplied document. Include mandatory actions, limits, operating conditions, warnings, test criteria, and maintenance requirements. Use a table with: ID, requirement, value/condition, source section, exact evidence quote, and uncertainty level. Do not convert recommendations into requirements. Do not fill gaps using general knowledge.

Your group's best catch: ____________________

Why it matters: ____________________

Mission 4: Hallucination Hunt

Ask These Questions

1. What is the required operating temperature range?
2. Under what condition must the device enter safe mode?
3. Does the document guarantee waterproof operation during full submersion?
4. How long will the battery last under every operating condition?
5. Who may authorise installation near a public warning system?
6. **What international standard certifies this product?** (Trick question)

Verification Prompt

Prompt

Recheck your previous answers using only the supplied source. For each answer, provide the section and exact evidence quote. Label: Supported, Partially Supported, Unsupported, or Contradicted. If the source does not contain the answer, state "not specified". Explain any earlier overstatement.

Hunt Score

- Supported claims correctly verified: ______
- Unsupported claims caught: ______
- Contradictions caught: ______
- Invented details caught: ______

Mission 5: Engineer's Reality Check

Three risks:

1. ______
2. ______
3. ______

One safeguard: ______
Accountability statement: As the engineer, I remain responsible for ______

Mission 6: Ed Discussion Exchange

Your post must include:

1. Your best technical-reading prompt
2. What you changed and why
3. One likely AI failure mode

Your reply must include:

1. One way to make their prompt more precise
2. One source-verification step
3. One ethical or safety consideration

Do not post personal data, private documents, or full AI chat logs.

Mission 7: Reading Memo Sprint

Use the memo template in Section 11. Include only claims you checked. Cite sources. Distinguish facts from assumptions. End with where you did not trust the AI.

AI Prompt Pack

Replace bracketed text before use. Always use `https://genai.hku.hk`.

1. **Quick Technical Summary**

   > Summarise the supplied technical document in five bullet points for a Year 2 engineering student. Cover: purpose, components, operating conditions, major requirements, safety issues. Cite a source section and evidence quote for every point. If absent, write "not specified". Do not infer.

2. **Requirement Extraction**

   > Extract all explicit requirements and constraints. Separate mandatory requirements, recommendations, assumptions, and background. Table format: requirement, value/condition, source section, evidence quote, confidence. Do not convert "may" or "should" into "must".

3. **Parameters and Units**

   > List every numerical parameter, limit, tolerance, duration, capacity, frequency, temperature, voltage, current, distance, and unit. Include the condition under which each applies. Cite section and evidence. Flag missing units or unclear conditions.

4. **Assumption Identification**

   > Identify assumptions stated or implied. Separate explicit assumptions from interpretations. Cite evidence. Label unsupported inferences as "inference requiring confirmation".

5. **Risk and Safety Identification**

   > Identify hazards, warnings, failure modes, misuse risks, and safety controls. Cite source section and evidence. Do not invent risk ratings. Flag any risk where severity or mitigation is not specified.

6. **Ambiguity and Missing Information**

   > Find ambiguous, incomplete, or untestable statements. Explain what is missing. Write one clarification question per issue. Cite source section.

7. **Terminology Explanation**

   > List acronyms and specialist terms. Explain using the document's wording where possible. Cite source. If undefined in the document, label as "general background".

8. **Source-Supported Q&A**

> Answer using only the supplied document: "[question]". One-sentence answer, then source section and exact quote. State "not specified" if unanswerable. Do not use outside knowledge.

9. **Verification Audit**

> Audit these AI-generated claims against the source: "[paste claims]". Label each: Supported, Partially Supported, Unsupported, or Contradicted. Provide section and evidence. Correct inaccuracies without adding new information.

10. **Engineering Checklist**

> Convert the document into a practical action checklist. Sections: before installation, installation, operation, testing, maintenance, emergency. Cite source beside every item. Include only source-supported actions.

11. **Challenge the AI's Own Answer**

> Act as a sceptical reviewer. Challenge your previous answer. Identify overstatements, missing conditions, wrong units, unsupported assumptions, or invented details. Recheck against the document with exact evidence.

12. **Questions for Supervisor**

> Generate ten questions an engineer should ask the supervisor/client before implementation. Focus on ambiguities, missing criteria, operating conditions, responsibilities, and safety. Cite the source that creates each question.

Discussion Prompts

Choose three to discuss:

1. When is AI most useful for reading technical documents?
2. When is AI dangerous or inappropriate for this task?
3. Which documents should never be blindly summarised?
4. What should you do if AI and the source disagree?
5. Who is responsible if an AI interpretation causes a design mistake?

One-Minute Decision

A document contains a warning that conflicts with an AI-generated checklist. You should:

A. Follow the checklist because it is clearer.
B. Average the two interpretations.
C. Stop, return to the source, document the conflict, and seek authorised clarification.
D. Ask the AI to choose.

What is you answer ?

Technical Document

SCFN-2 Smart Campus Flood Monitoring Node

Installation, Operation and Preliminary Test Note
Document status: Training draft, Revision 0.7
Fictional document for AILT9018

1. Purpose and Scope

The SCFN-2 is a battery-powered monitoring node that measures water level and environmental conditions near low-lying campus walkways. It sends readings to a campus gateway for facilities staff review. The node is **not** a certified emergency-warning device and must not be used as the sole basis for evacuation, road closure, or public-safety decisions.

This note covers installation, routine operation, communications, battery management, and preliminary acceptance testing. It does not cover structural design calculations, detailed cyber-security procedures, or approval for connection to public warning systems.

The system includes: ultrasonic water-level sensor, temperature sensor, rechargeable battery pack, solar charging panel, low-power controller, and wireless communication module.

2. Intended Operating Conditions

- Ambient temperature: **−10°C to 50°C**
- Relative humidity: up to **95% non-condensing**
- Enclosure: rated **IP65** when cable glands and access cover are correctly installed (dust and water jets; not continuous submersion)
- Ultrasonic sensor range: **0.20 m to 3.00 m** below sensor face; readings outside this range shall be reported as out-of-range

Design assumptions: clear downward view of water surface; mounting structure remains stable during heavy rain. Performance may be reduced by foam, floating debris, strong wind, angled water surfaces, or objects in the sensing path.

The solar panel should receive “adequate” daylight. No minimum daily solar exposure is defined in this note.

3. Installation Requirements

- Sensor face: at least **0.50 m above the highest expected water level**
- Sensor orientation: vertically downward within ±**5°**
- Mounting bracket must not obstruct sensor field of view
- Enclosure: mounted above local flood level where “reasonably practical” (no approval method or minimum height defined)
- All cable glands tightened; unused entries sealed; access cover closed with all four fasteners
- Record: coordinates, sensor height, device ID, and installation photographs
- Any sensor height change triggers a new reference-level measurement

WARNING 1: Electrical and Battery Hazard

Do not open the enclosure during heavy rain or while in floodwater. Disconnect battery before modifying wiring. Report damaged or swollen battery packs to the laboratory supervisor.

Installation within 2 m of an exposed power cable or electrical distribution cabinet requires a site-specific safety review. This note does not specify who approves that review.

4. Measurement and Reporting

- Normal: measure every **60 seconds**, transmit every **5 minutes**
- Rapid-reporting trigger: water level rises **>0.15 m within 5 minutes** → transmit every **30 seconds** for at least 10 minutes

Each message includes: device ID, timestamp, measured distance, calculated water level, battery voltage, signal quality, sensor status code.

Clock synchronised with gateway at least every 24 hours. If sync fails, timestamp marked as uncertain.

A "high water" message may be generated at a site-specific threshold (to be confirmed by facilities team). Until confirmed and documented, high-water message shall not trigger an automatic public alarm.

5. Power and Safe-State Behaviour

- Battery: **12.8 V nominal**, **6 Ah capacity**
- Low-battery warning: voltage below **11.5 V** for three consecutive measurements
- Safe mode: voltage below **10.8 V** — rapid reporting disabled, measurement every 10 minutes, one transmission attempt per hour
- Estimated operation without solar: approximately **6 days** (assumes full charge, 25°C, successful transmissions, no rapid-reporting events; not a guaranteed duration)

WARNING 2: Public-Safety Limitation

A missing message does not prove water level is safe. Communication failure, battery depletion, sensor blockage, or gateway failure may prevent an alert. Safety decisions shall use additional observations or approved monitoring systems.

6. Communications and Data Handling

Uses campus low-power wireless network. Data encrypted during transmission (algorithm, key management, and retention period not identified in this note).

Device identifiers shall not contain personal information. Installation photographs should avoid identifiable images of passers-by. Only "authorised project staff" may access detailed records (definition to be confirmed by supervisor).

Students shall not upload real location records, network credentials, unpublished datasets, or identifiable photographs to public AI services.

7. Acceptance Test

Before deployment, each node shall:

1. Confirm device ID matches project log
2. Confirm sensor height recorded to nearest **10 mm**
3. Test at two known distances; each reading within ±**20 mm**
4. Trigger simulated rapid-rise event; confirm 30-second reporting
5. Disconnect gateway; confirm local storage continues
6. Verify uncertain timestamp after sync failure
7. Inspect enclosure, cable glands, cover, battery, mounting

A node failing a mandatory check shall not be deployed until cause is recorded and corrective action completed. Who may approve a deviation from ±20 mm is not identified.

Preliminary test: three prototypes, sheltered outdoor area, 48 hours. No heavy-rain, submersion, prolonged network-loss, extreme-temperature, or six-day battery test was performed.

8. Maintenance and Professional Responsibility

Inspect sensor face and bracket every three months and after severe weather. Remove debris without changing alignment. Review battery on repeated low-voltage messages.

AI tools may help organise the document or extract candidate requirements. AI output shall be checked against the current approved revision. AI shall not replace site inspection, authorised approval, engineering review, or professional judgement.

A reviewer preparing a checklist remains responsible for accurate representation of requirements, warnings, limits, and unresolved questions. Unclear requirements shall be referred to the project supervisor.

Templates

Technical Summary Template

Document title: ____________________

Purpose (one sentence): ____________________

System covered: ____________________

Three key technical points:

1. ____________________ Source: ____________________
2. ____________________ Source: ____________________
3. ____________________ Source: ____________________

Main safety concern: ____________________

Important information not specified: ____________________

Requirement Extraction Table

ID	Requirement	Value	Section	Evidence quote	Confidence	Status
R1					H / M / L	
R2					H / M / L	
R3					H / M / L	
R4					H / M / L	
R5					H / M / L	
R6					H / M / L	

Status options: Supported / Partly Supported / Unsupported / Contradicted

AI Output Verification Table

AI claim	Section	Evidence found	Verdict	Action

Hallucination Tracker

Suspicious claim	Problem type	Why risky	Evidence check	Decision

Most serious AI mistake found: ____________________

What made it hard to notice: ____________________

Engineering Risk Checklist

Mark each: **Checked** / **Concern** / **N/A**

Check	Status	Evidence or action
Safety warnings identified		
Numerical limits checked		
Units checked		
Operating conditions checked		
Requirements separated from recommendations		
Assumptions labelled		
Ambiguities identified		
Test limitations identified		
Information gaps identified		
Privacy and confidentiality considered		
Human approval responsibilities identified		
AI claims traced to source		

AI-Assisted Technical Reading Memo

Document: ____________________
Reviewer: ____________________
Date: ____________________
Revision checked: ____________________

1. Verified Summary (3–5 sentences)

2. Five Key Requirements or Constraints

1. ____________ Source: ____________
2. ____________ Source: ____________
3. ____________ Source: ____________
4. ____________ Source: ____________
5. ____________ Source: ____________

3. Three Risks, Ambiguities, or Missing Details

1. ____________________
2. ____________________
3. ____________________

4. Two Verification Checks Performed

1. Claim: ____________________
   Evidence/result: ____________________
2. Claim: ____________________
   Evidence/result: ____________________

5. Recommended Next Action

6. Ethical or Professional Responsibility Reminder

7. How AI Helped Me, and Where I Did Not Trust It

AI helped me by ____________________
I did not trust it when ____________________

Final Deliverable

Your memo is complete when:

- Every technical claim traces to the document
- Numbers include units and conditions
- Assumptions are not presented as facts
- Missing information is clearly labelled
- AI-generated wording has been manually checked

Exit Ticket

Complete these three statements:

1. **One AI prompt I will reuse for technical reading is...**

2. **One AI mistake I learned to watch for is...**

3. **One responsibility I have as an engineer using AI is...**

Boss Level Challenge (Optional)

If your group finishes early:

1. Ask the AI to generate a complete installation checklist from the SCFN-2 document.
2. Try to break it by finding:
   - A missing requirement
   - A recommendation wrongly stated as a requirement
   - A missing condition
   - An invented approval process
   - A number with the wrong unit
   - A weakened warning
3. Give the AI your evidence and ask it to correct the checklist.
4. Compare the first and revised versions.

Most dangerous checklist error: ____________________
Source evidence: ____________________
Why an engineer should care: ____________________

Completion Checklist

- ☐ Used https://genai.hku.hk for AI-assisted technical reading
- ☐ Tested at least two prompts
- ☐ Extracted at least five requirements from the document
- ☐ Verified at least three AI claims against the source
- ☐ Identified at least one hallucination or unsupported claim
- ☐ Checked numbers, units, conditions, and warnings
- ☐ Posted one Ed Discussion contribution
- ☐ Responded to one classmate
- ☐ Written one AI-Assisted Technical Reading Memo
- ☐ Reflected on ethical responsibility when using AI

Professional Reality Check

AI may help you locate, organise, compare, and question technical information. It does not remove your responsibility to protect confidentiality, check the source, recognise uncertainty, seek authorised clarification, and make safe engineering decisions.